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Introduction to Docker

Docker is a containerization platform that allows developers to package an application together with its runtime dependencies. The resulting package can run consistently across development, testing, and production environments.

A container is an isolated process that shares the host operating system kernel while keeping the application's files, libraries, and configuration separated from other containers. This makes containers lighter than traditional virtual machines for many workloads.

Sample metadata context: section 1 of 5.

Images and Containers

A Docker image is an immutable template used to create containers. Images are commonly built from a Dockerfile, which describes the base image, application files, dependencies, environment configuration, and commands required to run the application.

A container is a running instance of an image. Multiple containers can be created from the same image, allowing teams to reproduce the same application environment across machines.

Sample metadata context: section 2 of 5.

Docker in Backend Development

Backend developers commonly use Docker to run APIs, databases, message brokers, caches, and other supporting services. A Spring Boot or FastAPI application can be packaged into an image and started without manually installing its runtime dependencies on the target machine.

Docker Compose can describe multiple services in one configuration. For example, an application stack may contain a FastAPI service, PostgreSQL database, Redis cache, and RabbitMQ broker connected through a shared network.

Sample metadata context: section 3 of 5.

Benefits and Tradeoffs

Containers improve portability, reproducibility, dependency isolation, and deployment consistency. They also make it easier to create repeatable development and testing environments.

Docker does not remove operational complexity. Teams still need to manage image security, secrets, networking, persistent storage, logging, monitoring, resource limits, and production deployment strategies.

Sample metadata context: section 4 of 5.

Summary

Docker packages applications into images and executes those images as containers. Images provide repeatable application environments, while containers provide isolated runtime instances.

For backend engineering, Docker is especially useful for local development, automated testing, deployment pipelines, and running applications together with infrastructure services. Understanding images, containers, volumes, networks, Dockerfiles, and Docker Compose provides a strong foundation for container-based development.

Sample metadata context: section 5 of 5.